Thermal Entropy Slack and Certified Scalar Optimization for Generalized Amplitude Damping

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M559

October 9, 2026

Abstract

We examine the finite-dimensional block mechanism in the source manuscript *Classical capacity and entropy inequalities for generalized amplitude damping*. Its zero-block entropy inequality is retained as a named analytic input, not replaced by positivity or a finite matrix test. Applying classical strong entropy concavity to the source's exact two-endpoint thermal mixture retains the additional term $\beta\eta \left\| XX^* - YY^* \right\|_1^2 / (2U_0U_1)$. We prove its rank-independent coefficient, singular conventions, and uniform sharpness. For arbitrary finite-dimensional partner channels this gives a quantitative deficit for pure signal ensembles; for an identity partner it bounds their average input linear entropy. Independently, the channel's scalar objective has a nonnegative polynomial series with explicit tails. Rational Bernstein subdivision and a strict-concavity estimate give value and optimizer certificates, including a worked rational example. All entropies, branch weights, and physical norms are specified. The source's unrestricted capacity and universal partner statements are not independently certified by these local results. Contribution assessment: C3/5, the explicit thermal slack, conditional near-product ensemble bound, and positive-series scalar certificates are notable source-specific refinements, not a new global capacity theorem.

Contents

1 Channel, conventions, and the source dependency boundary **2**

2 A sharp quantitative remainder in the thermal mixture **3**

3 Channel-specific branch and ensemble deficits **4**

4 A positive polynomial series for the scalar optimization **7**

5 Physical coordinates and original Ram proof connections **9**

6 Architecture, current literature, and comparison **9**

7 Verification scope and conclusion **11**

1 Channel, conventions, and the source dependency boundary

All Hilbert spaces in this report are finite-dimensional complex spaces with their physical inner products. The adjoint is $*$, the trace norm is $\|D\|_1 = \operatorname{Tr}\sqrt{D^*D}$, and

$$\mathcal{S}(P) = -\operatorname{Tr} P \ln P, \qquad 0 \ln 0 = 0, \qquad h(t) = -t \ln t - (1-t)\ln(1-t).$$

Entropy is in nats unless a factor $1/\ln 2$ is written. For a positive matrix P of arbitrary trace, $\mathcal{S}$ has this same definition; it is not the entropy of its normalization. Density matrices have trace one. Write

$$\mathcal{A}_{\gamma,\nu}\begin{pmatrix} 1-p & z \\ \bar{z} & p \end{pmatrix} = \begin{pmatrix} 1-t(p) & \sqrt{1-\gamma}\, z \\ \sqrt{1-\gamma}\,\bar{z} & t(p) \end{pmatrix}, \qquad t(p) = (1-\gamma)p + \gamma\nu. \tag{1}$$

Here $\gamma, \nu, p \in [0,1]$, $|z|^2 \le p(1-p)$, and ν is the stationary *excited* population. In particular, it is not silently replaced by the ground-state convention. Set

$$g(u) = h\left(\frac{1+\sqrt{1-4u}}{2}\right), \quad 0 \le u \le \tfrac{1}{4}, \tag{2}$$

$$v(p) = \gamma\nu(1-\nu) + \gamma(1-\gamma)(p-\nu)^2, \qquad e(p) = g(v(p)), \qquad F(p) = h(t(p)) - e(p). \tag{3}$$

The determinant of the output of a pure qubit with population p is $v(p)$. Thus $e(p)$ is its output entropy, independently of its phase. Define the scalar candidate $C_1 = (\max_{[0,1]} F)/\ln 2$.

For a finite ensemble $\mathcal{E} = \{q_i, \rho_i\}$ and a channel $\mathcal{M}$,

$$I_{\mathcal{M}}(\mathcal{E}) = \frac{\mathcal{S}(\sum_i q_i \mathcal{M}\rho_i) - \sum_i q_i \mathcal{S}(\mathcal{M}\rho_i)}{\ln 2}, \qquad \chi(\mathcal{M}) = \sup_{\mathcal{E}} I_{\mathcal{M}}(\mathcal{E}).$$

These are one-shot Holevo quantities in bits, not accessible information for a prescribed measurement. The regularized unassisted capacity uses $\sup_n \chi(\mathcal{M}^{\otimes n})/n$ and a coding theorem. Arbitrary entangled block inputs and collective decoding are allowed in the source. No assertion about separate-measurement decoding is made here.

1.1 What is imported and what is proved below

The source [1] proves a zero-block inequality, then a thermal block inequality, conditional output bounds, and the scalar capacity formula. Its partner argument quantifies over *every* finite CPTP map, not just another damping channel. The precise analytic input used in our thermal deductions is the following statement, called **Z**:

$$\mathcal{S}(WW^*) \ge x\mathcal{S}(AA^*/x) + y\mathcal{S}(BB^*/y) + z\mathcal{S}(CC^*/z) + g(yz), \quad W = \begin{pmatrix} A & B \\ C & 0 \end{pmatrix}. \tag{4}$$

The rectangular blocks have compatible equal block sizes, $x = \operatorname{Tr} AA^*$, $y = \operatorname{Tr} BB^*$, $z = \operatorname{Tr} CC^*$, $x + y + z = 1$; zero-weight terms are omitted. Rectangular padding by zeros does not change nonzero spectra. Statement Z is the source's zero-block theorem in `block.tex`. Its full written proof was freshly read; we do not turn that reading into an independent certification of all global quantum-information claims. Every use of Z below is explicit. The strong-concavity lemma and the scalar-series and optimization results are proved without Z.

The source proof of Z uses an equal-trace entropy/log-determinant integral, joint convexity of

$$G_Z(D,E) = \ln\det(D + ZE^{-1}Z^*) - \ln\det D, \qquad D, E > 0,$$

sign averaging after a singular-value decomposition, and a homogeneous concave scalar deficit. In particular, PSD of WW^* is only a domain condition, not a proof of (4). The regularizers $I + rP$ in that proof remain invertible even if P is singular. The alternative source Hessian proof and all singular endpoint discussions were also read.

2 A sharp quantitative remainder in the thermal mixture

We first supply a self-contained form of a classical entropy fact. It is equivalent in this application to applying quantum Pinsker in nats; Pinsker's constant is not claimed as a new result [5].

Lemma 2.1 (Strong entropy concavity)**.** *For density matrices A, B on the same space and $w \in [0,1]$,*

$$\mathcal{S}(wA + (1-w)B) - w\mathcal{S}(A) - (1-w)\mathcal{S}(B) \geq \frac{w(1-w)}{2} \|A - B\|_1^2 . \tag{5}$$

The constant is independent of dimension and rank.

Proof. First take $A, B > 0$, put $D = A - B$ and $\rho_s = B + sD$. Differentiating $-\operatorname{Tr} \rho_s \ln \rho_s$ twice, in an eigenbasis of ρ_s, gives

$$-\frac{d^2}{ds^2}\mathcal{S}(\rho_s) = \sum_{ij} |D_{ij}|^2 \frac{\ln \lambda_i - \ln \lambda_j}{\lambda_i - \lambda_j} \geq \sum_{ij} \frac{2|D_{ij}|^2}{\lambda_i + \lambda_j}.$$

The divided difference equals the integral of $((1-t)\lambda_j + t\lambda_i)^{-1}$ over $[0,1]$; convexity of the reciprocal proves the inequality, including equal eigenvalues by limits. Choose a Hermitian unitary J agreeing with $\operatorname{sign} D$ on its nonzero eigenspaces. Weighted Cauchy–Schwarz yields

$$(\operatorname{Tr} JD)^2 \leq \left(\sum_{ij} \frac{2|D_{ij}|^2}{\lambda_i + \lambda_j} \right) \left(\sum_{ij} \frac{\lambda_i + \lambda_j}{2} |J_{ij}|^2 \right).$$

The last factor is $\operatorname{Tr} \rho_s J^2 = 1$ and $\operatorname{Tr} JD = \|D\|_1$. Subtracting the quadratic $\|D\|_1^2 \, s(1-s)/2$ from the entropy chord gap leaves a concave function vanishing at both endpoints. It is nonnegative, proving (5). For singular matrices apply this argument to $(1-\epsilon)A + \epsilon I/d$ and $(1-\epsilon)B + \epsilon I/d$, and use finite-dimensional entropy continuity and $\|(1-\epsilon)D\|_1 \to \|D\|_1$. □

Let $X, Y \in \mathbb{C}^{d \times r}$ satisfy $\operatorname{Tr} XX^* = \operatorname{Tr} YY^* = 1$ and write $A = XX^*$, $B = YY^*$. For nonnegative $\alpha, \beta, \eta, \delta, K$ assume

$$\alpha + \beta + \eta + \delta = 1, \qquad \alpha\delta - \beta\eta = K^2. \tag{6}$$

The physical density matrix and its row weights are

$$T = \begin{pmatrix} \alpha A + \beta B & KXY^* \\ KYX^* & \eta A + \delta B \end{pmatrix}, \quad U_0 = \alpha + \beta, \quad U_1 = \eta + \delta, \quad u = U_0 U_1 - K^2 = \beta U_1 + \eta U_0. \tag{7}$$

In particular $0 \leq u \leq 1/4$. The common right-hand space of X, Y is essential: the off-diagonal block is the actual XY^*, not an independently selected contraction.

Theorem 2.2 (Thermal slack, conditional on Z)**.** *Under* (6)*, statement Z implies*

$$\mathcal{S}(T) \geq (\alpha + \eta)\mathcal{S}(A) + (\beta + \delta)\mathcal{S}(B) + g(u) + c_T \|A - B\|_1^2, \quad c_T = \begin{cases} \dfrac{\beta\eta}{2U_0U_1}, & U_0U_1 > 0, \\ 0, & U_0U_1 = 0. \end{cases} \tag{8}$$

All ranks and all boundary coefficients are permitted. The numerical factor $1/2$ in the displayed coefficient cannot be uniformly enlarged over this class.

Proof. Suppose first $u > 0$ and $U_0U_1 > 0$. Use the source's two coefficient quadruples, in the order $(\alpha, \beta, \eta, \delta)$,

$$(K^2/U_1, u/U_1, 0, U_1), \qquad (U_0, 0, u/U_0, K^2/U_0), \quad w_1 = \frac{\beta U_1}{u}, \quad w_2 = \frac{\eta U_0}{u}.$$

They are nonnegative, sum to one, have the same determinant relation, and $w_1 + w_2 = 1$. Let their matrices be T_1, T_2, retaining the same off-diagonal KXY^*. Direct substitution gives $T = w_1T_1 + w_2T_2$. For T_1 use the factor

$$\begin{pmatrix} \sqrt{K^2/U_1}X & \sqrt{u/U_1}Y \\ \sqrt{U_1}Y & 0 \end{pmatrix};$$

for T_2 use $\begin{pmatrix} \sqrt{U_0}X & 0 \\ \sqrt{K^2/U_0}Y & \sqrt{u/U_0}X \end{pmatrix}$. For the latter, permute its two block rows before using Z. These factors give the source's entropy bound at each endpoint with the same $g(u)$. Their difference, with $D = B - A$, is exactly

$$T_1 - T_2 = \operatorname{diag}((u/U_1)D, (u/U_0)D), \qquad \|T_1 - T_2\|_1 = \frac{u}{U_0U_1}\|D\|_1. \tag{9}$$

Thus Lemma 2.1 adds

$$\tfrac{1}{2}w_1w_2\|T_1 - T_2\|_1^2 = \frac{\beta\eta}{2U_0U_1}\|D\|_1^2.$$

The weighted endpoint entropy coefficients recombine to those in (8).

If $u = 0$ and both row weights are positive, then $\beta = \eta = 0$. The remaining factor is the column $(\sqrt{\alpha}X, \sqrt{\delta}Y)^T$. Its nonzero spectrum equals that of $\alpha X^*X + \delta Y^*Y$. Entropy concavity and equality of the nonzero spectra of XX^* and X^*X give the required bound, with $g(0) = c_T = 0$. If a row weight vanishes, $K = 0$ and the other row is an ordinary mixture of A, B; again entropy concavity suffices. These direct cases avoid assigning a value to a $0/0$ endpoint decomposition.

For sharpness take $\alpha = \beta = \eta = \delta = 1/4$, $K = 0$, $A = \operatorname{diag}(1/2 + t, 1/2 - t)$, $B = \operatorname{diag}(1/2 - t, 1/2 + t)$, $0 < t < 1/2$, and $X = \sqrt{A}$, $Y = \sqrt{B}$. Then $T = I_4/4$ and its gap above the three source terms is $\ln 2 - h(1/2 + t) = 2t^2 + O(t^4)$. The added term is exactly $(1/8)(4t)^2 = 2t^2$. A larger universal factor fails as $t \downarrow 0$. □

Remark 2.3 (Necessary is not sufficient)**.** If $\beta\eta > 0$, equality in the original thermal bound forces $A = B$. This is only a necessary condition. Equality also needs equality in the two zero-block bounds. The source's sufficient example $X = Y$ is stronger than $XX^* = YY^*$; the cross block XY^* still matters. No complete equality classification is claimed.

3 Channel-specific branch and ensemble deficits

Put $s = 1 - \gamma$, $a = 1 - \gamma\nu$, $b = \gamma(1 - \nu)$, $c = \gamma\nu$, $d = 1 - \gamma(1 - \nu)$, so $a + c = b + d = 1$ and $ad - bc = s$. Let $\mathcal{N}$ be any finite CPTP partner, with a common Kraus family V_ℓ. For a unit vector

$$|\psi\rangle = \sqrt{1 - p}|0\rangle|\psi_0\rangle + \sqrt{p}|1\rangle|\psi_1\rangle, \qquad \|\psi_0\| = \|\psi_1\| = 1,$$

the branch vectors need not be orthogonal. At $p = 0$ or 1, choose the unused unit vector arbitrarily. Take X, Y to have columns $V_\ell\psi_0, V_\ell\psi_1$. Then

$$A = \sigma_0 = \mathcal{N}(|\psi_0\rangle\langle\psi_0|), \quad B = \sigma_1 = \mathcal{N}(|\psi_1\rangle\langle\psi_1|), \quad XY^* = \mathcal{N}(|\psi_0\rangle\langle\psi_1|).$$

Trace preservation gives the required unit traces. Substituting

$$(\alpha, \beta, \eta, \delta) = (a(1-p), bp, c(1-p), dp), \qquad K = \sqrt{sp(1-p)} \tag{10}$$

in (7) produces the actual output, with $U_1 = t(p)$ and $u = v(p)$, not just a matrix with matching eigenvalues.

Corollary 3.1 (Resolved branch-information penalty)**.** *Conditional on Z, for every such $\mathcal{N}, \psi$,*

$$\mathcal{S}((\mathcal{A}_{\gamma,\nu} \otimes \mathcal{N})(\psi)) \geq e(p) + (1-p)\mathcal{S}(\sigma_0) + p\mathcal{S}(\sigma_1) + c(p) \left\|\sigma_0 - \sigma_1\right\|_1^2, \tag{11}$$

$$c(p) = \frac{\gamma^2 \nu(1-\nu)p(1-p)}{2t(p)(1-t(p))} \quad \textit{when } 0 < t(p) < 1, \tag{12}$$

and $c(p) = 0$ if $t(p) \in \{0, 1\}$.

The assertion follows by substitution in Theorem 2.2. For $\gamma > 0$, $0 < \nu < 1$, and $0 < p < 1$, its coefficient is strictly positive. It vanishes at pure thermal endpoints or undamped $\gamma = 0$. For $\gamma = 1$ and $0 < \nu < 1$, it reduces to $p(1-p)/2$.

3.1 Scalar curvature needed for an ensemble certificate

The source already proves convexity of e, a fact we retain rather than count as a new result. Here is its short scalar verification. For $z = \sqrt{1-4u}$ and $L = \operatorname{atanh}(z)/z$ in the open interval,

$$g'(u) = 2L, \qquad ug''(u) + \tfrac{1}{2}g'(u) = \frac{L-1}{z^2} \geq 0.$$

Consequently $x \mapsto g(x^2)$ is increasing and convex for $0 \leq x \leq 1/2$, extending continuously to the endpoints. The square root of $v(p)$ is the Euclidean norm of the affine vector $(\sqrt{\gamma\nu(1-\nu)}, \sqrt{\gamma s}(p-\nu))$. Composition with this norm proves convexity of e.

Proposition 3.2 (Strict scalar curvature)**.** *For $0 \leq \gamma < 1$ and every $\nu \in [0, 1]$, F is strongly concave on $[0, 1]$ with modulus $\mu = 4(1-\gamma)^2$. It has a unique maximizer $p_* \in (0, 1)$ and*

$$F(p_*) - F(p) \geq 2s^2(p - p_*)^2. \tag{13}$$

For $p \in (0, 1)$ a derivative certificate also gives

$$|p - p_*| \leq \frac{|F'(p)|}{\mu}, \qquad 0 \leq F(p_*) - F(p) \leq \frac{|F'(p)|^2}{2\mu}. \tag{14}$$

For $\gamma = 1$, $F \equiv 0$ and neither uniqueness nor division by μ is asserted.

Proof. Since $h''(t) = -1/[t(1-t)] \leq -4$ in $(0, 1)$, $h(t(p))$ is $4s^2$-strongly concave. Subtracting convex e preserves this modulus; continuity extends the chord inequality to the closed interval. The two endpoints of F are zero. For $s > 0$ and $0 < p < 1$, the two phase-opposite pure signals have distinct channel outputs, since their off-diagonal entries are $\pm\sqrt{sp(1-p)}$. Their mean is diagonal. Strict entropy concavity (or Lemma 2.1) shows $F(p) > 0$. Compactness and strong concavity therefore give a unique interior maximizer with $F'(p_*) = 0$. The strong tangent inequality gives (13), the derivative is strongly decreasing, and

$$F(p_*) - F(p) \leq F'(p)(p_* - p) - \tfrac{\mu}{2}(p_* - p)^2 \leq \frac{|F'(p)|^2}{2\mu}.$$

This proves the remaining claims. When $\gamma = 1$, $t = \nu$ and $e = h(\nu)$, proving the stated exception. □

For interior p and $0 < \gamma < 1$ one may use

$$F'(p) = s \ln \frac{1 - t(p)}{t(p)} - 2\gamma s(p - \nu) g'(v(p)). \tag{15}$$

At $v = 1/4$, $g' = 2$ by a limit. The singular $g'(0)$ must not be substituted as a finite number: the undamped case is simply $F = h$, and boundary populations use continuous one-sided values as needed. A verified derivative interval $[d - \epsilon, d + \epsilon]$ may replace $|F'(p)|$ by $|d| + \epsilon$ in (14). A floating stationarity residual with no enclosure is not such a certificate.

Theorem 3.3 (Pure-ensemble additivity deficit, conditional on Z)**.** *Let $\mathcal{E} = \{q_i, \psi_i\}$ be any finite ensemble of pure inputs to $\mathcal{A}_{\gamma,\nu} \otimes \mathcal{N}$, with the preceding branch populations p_i, outputs σ_{0i}, σ_{1i}, and $\bar{p} = \sum_i q_i p_i$. Put*

$$J_e = \sum_i q_i e(p_i) - e(\bar{p}) \geq 0, \quad D_i = \|\sigma_{0i} - \sigma_{1i}\|_1, \quad \Delta = \ln 2 \, [C_1 + \chi(\mathcal{N}) - I_{\mathcal{A}_{\gamma,\nu} \otimes \mathcal{N}}(\mathcal{E})].$$

For $\gamma < 1$,

$$\Delta \geq 2s^2(\bar{p} - p_*)^2 + J_e + \sum_i q_i c(p_i) D_i^2. \tag{16}$$

For $\gamma = 1$ the same inequality holds with the first term omitted.

Proof. Let $\tau = \sum_i q_i[(1 - p_i)\sigma_{0i} + p_i \sigma_{1i}]$. It is the actual mean partner marginal; no averaging of transition laws or independent replacement of cross blocks is made. Subadditivity and dephasing on the qubit imply

$$\mathcal{S}(\overline{(\mathcal{A}_{\gamma,\nu} \otimes \mathcal{N})\psi}) \leq h(t(\bar{p})) + \mathcal{S}(\tau).$$

Average (11) and subtract. The remaining expression $\mathcal{S}(\tau) - \sum_i q_i[(1 - p_i)\mathcal{S}(\sigma_{0i}) + p_i \mathcal{S}(\sigma_{1i})]$ is the Holevo information in nats of the genuine partner input ensemble with weights $q_i(1 - p_i), q_i p_i$, hence is at most $\ln 2 \, \chi(\mathcal{N})$. It follows that

$$\ln 2 \, I_{\mathcal{A}_{\gamma,\nu} \otimes \mathcal{N}}(\mathcal{E}) \leq F(\bar{p}) + \ln 2 \, \chi(\mathcal{N}) - J_e - \sum_i q_i c(p_i) D_i^2.$$

Use Proposition 3.2, or $F = 0$ in the replacer case. □

This calculation does not use an independent global additivity theorem as a black box, but it does use Z through (11). If an ensemble is within ϵ bits of the candidate sum, each nonnegative term on the right of (16) is at most $\epsilon \ln 2$. For $p_i \in [\ell, 1 - \ell]$, $0 < \ell \leq 1/2$, $c(p_i) \geq 2\gamma^2 \nu(1 - \nu)\ell(1 - \ell)$; hence the total weight of those signals with $D_i \geq d_0 > 0$ is at most

$$\frac{\epsilon \ln 2}{2\gamma^2 \nu(1 - \nu)\ell(1 - \ell) d_0^2}, \qquad \gamma > 0, \quad 0 < \nu < 1.$$

This is a conditional structural bound, not a construction of an optimal code or a uniform bound as $\gamma\nu(1 - \nu) \to 0$.

Corollary 3.4 (Identity partner and near-product signals)**.** *For $\mathcal{N} = \mathrm{id}_d$, $\gamma > 0$, $0 < \nu < 1$, an ensemble within ϵ bits of $C_1 + \log_2 d$ obeys*

$$\sum_i q_i (1 - \operatorname{Tr} \rho_{A,i}^2) \leq \frac{\epsilon \ln 2}{4\gamma^2 \nu(1 - \nu)}. \tag{17}$$

Here $\rho_{A,i}$ is the qubit marginal of the pure input ψ_i. At zero deficit every positive-weight signal is a product vector.

Proof. For the identity partner, $D_i^2 = 4(1 - |\langle\psi_{0i}, \psi_{1i}\rangle|^2)$ and $L_i := 1 - \operatorname{Tr}\rho_{A,i}^2 = 2p_i(1-p_i)(1 - |\langle\psi_{0i}, \psi_{1i}\rangle|^2)$. Consequently $c(p_i)D_i^2 = \gamma^2\nu(1-\nu)L_i/[t(p_i)(1-t(p_i))] \geq 4\gamma^2\nu(1-\nu)L_i$. The endpoint populations have $L_i = 0$. Sum and apply Theorem 3.3. A pure bipartite state has a pure qubit marginal exactly when it is a product, proving the last assertion. □

For a general partner, $D_i = 0$ need not mean product input. For example, a Bell vector has orthogonal branches, but a replacer channel $\mathcal{N}(\rho) = \operatorname{Tr}(\rho)\omega$ gives both branch outputs ω and zero penalty. This example precludes interpreting (16) as an input-entanglement certificate for arbitrary partners. Pure-signal ensembles are essential to the stated linear-entropy interpretation; refinement of a mixed ensemble changes its signals.

4 A positive polynomial series for the scalar optimization

The objective F is already in the source. The following representation retains its positivity term by term and separates exact polynomial arithmetic from a proved infinite tail. It does not depend on Z.

Theorem 4.1 (Polynomial lower bounds and explicit tails)**.** *For every $\gamma, \nu, p \in [0,1]$, set*

$$Y(p) = (1-2t(p))^2, \qquad R(p) = 1 - 4v(p), \qquad R(p) - Y(p) = 4(1-\gamma)p(1-p).$$

Then $0 \leq Y \leq R \leq 1$ and

$$F(p) = \sum_{k=1}^{\infty} \frac{R(p)^k - Y(p)^k}{2k(2k-1)}, \qquad P_K(p) = \sum_{k=1}^{K} \frac{R(p)^k - Y(p)^k}{2k(2k-1)} \leq F(p). \tag{18}$$

The series converges uniformly over the entire parameter cube. For $K \geq 1$ and a cell $I \subset [0,1]$ on which $R(p) \leq r_I < 1$,

$$0 \leq F(p) - P_K(p) \leq B_K(r_I) := \frac{r_I^{K+1}}{(2K+2)(2K+1)(1-r_I)}. \tag{19}$$

A universally valid, slower alternative is $F - P_K \leq 1/(2K)$, even when $r_I = 1$. If $\gamma > 0$ and $0 < \nu < 1$, a uniform choice is $r_I = 1 - 4\gamma\nu(1-\nu) < 1$.

Proof. The pure output is a density matrix with determinant v, so $0 \leq v \leq 1/4$. Its diagonal product is $t(1-t)$ and its squared off-diagonal modulus is $sp(1-p)$, giving the identity $R - Y = 4sp(1-p)$ and the range claims. For $|x| < 1$, differentiating binary entropy gives

$$\frac{d}{dx}h((1+x)/2) = -\operatorname{atanh} x = -\sum_{j=0}^{\infty} \frac{x^{2j+1}}{2j+1}.$$

Integration from zero yields $h((1+x)/2) = \ln 2 - \sum_{k\geq 1} x^{2k}/[2k(2k-1)]$. Since the coefficients are summable, the identity extends uniformly to $[-1,1]$. Apply it with $x = 1-2t$ and $x = \sqrt{R}$ and subtract to get (18). The terms are nonnegative. For $k > K$, replace $2k(2k-1)$ by $(2K+2)(2K+1)$ and sum the geometric majorant $\sum_{k>K} r_I^k$, proving (19). Finally, $1/[2k(2k-1)] \leq 1/(2k^2)$ and $\sum_{k>K} k^{-2} \leq \int_K^\infty x^{-2}dx = 1/K$. This also proves the claimed cube-uniform convergence. The lower bound $v(p) \geq \gamma\nu(1-\nu)$ proves the last assertion. □

The series makes several source details transparent. For $\gamma = 1$, $R = Y$ identically and every $P_K = F = 0$, so one should return the exact zero value, not a spurious positive truncation error. If $\gamma = 0$, then $R = 1$ and $F = h(p)$; the universal tail remains valid. At $p = 0, 1$, again $R = Y$ and each term vanishes. No eigenvalue logarithm is evaluated at zero in this construction. For rational parameters, P_K has rational coefficients and degree at most $2K$.

Proposition 4.2 (Whole-interval value and optimizer enclosure)**.** *Let $I_j = [a_j, b_j]$ form a finite cover of $[0, 1]$. Express $P_K(a_j + (b_j - a_j)x) = \sum_{\ell=0}^{2K} \beta_{j\ell} B_\ell^{2K}(x)$, where $B_\ell^n(x) = \binom{n}{\ell} x^\ell (1-x)^{n-\ell}$; lower actual degrees may instead be used. Choose a valid tail bound b_j^{tail} from Theorem 4.1, and any $p_0 \in [0, 1]$. Then*

$$L := P_K(p_0) \le \max F \le U := \max_j \left(\max_\ell \beta_{j\ell} + b_j^{\mathrm{tail}} \right). \tag{20}$$

For $\gamma < 1$ the unique maximizer additionally satisfies

$$|p_* - p_0| \le \sqrt{\frac{U - L}{2(1-\gamma)^2}}. \tag{21}$$

All coefficients and bounds in (20) *can be exact rationals when the channel parameters, cells, and p_0 are rational and r_I is rational. Divide by a separately enclosed positive* $\ln 2$ *for bits.*

Proof. The Bernstein basis is nonnegative and sums to one, so its largest coefficient bounds the polynomial throughout the cell. To make the conversion explicit, if $P(x) = \sum_{k=0}^n \alpha_k x^k$, then

$$\beta_\ell = \sum_{k=0}^{\ell} \alpha_k \frac{\binom{\ell}{k}}{\binom{n}{k}}.$$

Indeed the identity $x^k = \sum_{\ell=k}^n [\binom{\ell}{k}/\binom{n}{k}] B_\ell^n(x)$ follows from the binomial theorem after factoring x^k. Add the proved tail on each cell and maximize. The lower bound is Theorem 4.1; (13) gives $2s^2|p_* - p_0|^2 \le F(p_*) - F(p_0) \le U - L$. □

Exact subdivision uses convex combinations of adjacent Bernstein coefficients (de Casteljau's algorithm). For fixed K, the hull overestimation tends uniformly to zero as cells shrink: after affine rescaling, all nonconstant power coefficients are $O(|I_j|)$ with constants depending on P_K, and the finite conversion matrix is fixed. First choose K to control the analytic tail, then refine cells to control the hull error. This proves convergence of the value enclosure, but does not give a parameter-uniform fast algorithm. In particular $r_I \uparrow 1$ slows the geometric tail, and $\gamma \uparrow 1$ weakens the optimizer bound even though the exact replacer objective is trivial. Finite sample maxima without either a hull or a derivative remainder do not enclose the maximum.

Example 4.3 (A rational two-cell certificate)**.** Take $\gamma = \nu = 1/2$. With $x = p - 1/2$ one obtains

$$R = \tfrac{1}{2} - x^2, \quad Y = x^2, \quad P_2 = \tfrac{13}{48} - \tfrac{13}{12}x^2 = \tfrac{13}{12}p(1-p).$$

In degree two, the Bernstein coefficients on $[0, 1/2]$ are $(0, 13/48, 13/48)$, and on $[1/2, 1]$ are $(13/48, 13/48, 0)$. A single cell would give the less useful upper hull $13/24$. Since $R \le 1/2$, the analytic tail is at most $B_2(1/2) = 1/120$. Thus exact rational arithmetic proves

$$\frac{13}{48} \le \max F \le \frac{67}{240}, \qquad \text{width } \frac{1}{120} \text{ nats.} \tag{22}$$

Here symmetry and strict concavity already place p_* exactly at $1/2$; this is a test of the value certificate, not a new discovery of the unital optimizer. Increasing K and refining the cells sharpens the enclosure with the same proof. The source's exact value is $\ln 2 - g(1/8)$, consistent with (22).

5 Physical coordinates and original Ram proof connections

The relevant original material is used at two precise interfaces, not as a substitute for quantum entropy theory. The physical-metric chapter [7] proves $G = J^*J$, orthogonal projection $JG^{-1}J^*$, metric adjoints, and the omitted-residual identity. In the present finite problem, if $X = JX_c$, $Y = JY_c$ in an injective nonorthogonal synthesis, then the isometry $U = JG^{-1/2}$ gives

$$XX^* - YY^* = U\,G^{1/2}(X_cX_c^* - Y_cY_c^*)G^{1/2}U^*,$$

and therefore

$$\|XX^* - YY^*\|_1 = \left\|G^{1/2}(X_cX_c^* - Y_cY_c^*)G^{1/2}\right\|_1, \qquad \operatorname{Tr} X_c^*GX_c = 1. \tag{23}$$

This follows because an isometry preserves nonzero singular values. For the whole thermal block the isometry is $\operatorname{diag}(U, U)$; consequently the block entropy is also preserved, with only extra zero eigenvalues outside the range. The coefficient matrix $X_cX_c^*$ without G is not the physical density matrix. These are classical coordinate facts, not the contribution score's basis. No infinite-dimensional frame or arbitrary partner result follows from a finite Gram matrix.

The original polynomial chapter [8], especially its power-to-Bernstein proof, hull enclosure, exact subdivision, and nonpolynomial-remainder discussion, supplies the finite arithmetic method in Proposition 4.2. Our channel-specific work is the positive series (18) and its proved tail, which make that method applicable to the entropy difference without replacing logarithms by an unjustified fitted polynomial. The chapter's obstacle and ODE examples were read but are not claimed as evidence about a quantum channel. Its external figures and cited works were not audited.

6 Architecture, current literature, and comparison

6.1 Complete source map, with retained obligations

The assigned public text has ten files. Their scientific roles are:

File in the build	Role and retained dependency
`paper.tex`	Model, operational statements, tensor/partner quantifiers, normalizations and source historical comparison.
`block.tex`	Entropy integral; joint log-determinant convexity; sign averaging; scalar concavity; zero-block theorem Z.
`thermal.tex`	Exact two-endpoint mixture and singular cases; the present extra concavity term is retained here.
`output.tex`	Common-Kraus partner blocks, convex e, conditional output bound, iteration over full tensor inputs.
`holevo.tex`	Mean-output upper bound, individual-output lower bound, phase-pair attainment and repeated-use conclusion.
`partners.tex`	Genuine conditional partner ensemble, universal partner Holevo/minimum-entropy claims and regularization.
`special.tex`	Thermal reflection, unital maximum and singular damping/replacer cases.
`alternative.tex`	Alternative scalar and matrix Hessian proofs; no change to the entropy integral or sign-average steps.
`bibliography.tex`, README	Public references and release metadata, not proof audits of the referenced papers.

The local derivation preserves the source's common Kraus cross block, both positive diagonal thermal contributions, scalar entropy correction, population conditioning, and arbitrary partner dimension. A numerical PSD test, a Schur complement, or agreement of a few output eigenvalues cannot supply Z or the operational coding theorem. We have not proved a new uniform statement about arbitrary block-length codes by finite tests.

6.2 Primary public status checked October 9, 2026

Khatri–Sharma–Wilde [2], arXiv:1903.07747v2 (July 8, 2020), provides capacity bounds and structural GAD analysis. Its primary abstract and version/publication metadata were checked, not its full proofs. Tang–Zhu–Bai–Wang [3], arXiv:2609.28592v1 (September 23, 2026), states arbitrary-partner results for qubit channels admitting a pure output. That scope covers ordinary damping but is not, by itself, all interior thermal channels. In fact, for $\gamma > 0$ and $0 < \nu < 1$, $v(p) \geq \gamma\nu(1-\nu) > 0$, so no pure input has a pure output, and mixing inputs cannot create one. Tang et al. also state a finite-block deficit for ordinary damping; no historical novelty for deficit methods is claimed here.

A material bibliographic update is necessary. The cached source cites Fang's v1 *A pretty-tight converse*. The current primary record [4] instead lists v2, October 6, 2026, *The classical capacity of generalized amplitude-damping channels and its strong-converse exponent*. Its abstract claims exact capacity and the strong-converse exponent throughout the parameter range through a triangular Schatten-norm route. We checked that abstract and version record, not the new proof. Thus the source's v1 comparison is not a complete description of the current literature; neither a remaining general capacity gap nor independent verification of v2 is asserted.

The repository's public Lean scope note [6] declares repeated-use GAD additivity and an operational/scalar capacity statement, but explicitly excludes arbitrary-different-partner additivity and separate-measurement decoding. This is a scope declaration, not a Lean execution or formal-proof audit by this report. Carlen's primary abstract [5] confirms the classical sharp quantum Pinsker context; Lemma 2.1 gives the particular proof needed here directly.

6.3 Advantages, costs, and further questions

The entropy remainder is independent of rank and dimension, survives singular matrices, and measures a physically specified branch-output distance. The identity-partner corollary adds quantitative information about near-optimal signals beyond a bare additive upper bound. Its cost is degeneracy at pure thermal parameters and the need to retain Z. The necessary equality condition does not classify all endpoint equality cases. For general partners, reversing channel contraction would require an additional, proved lower distinguishability bound on the particular input family; it cannot hold for replacer channels.

The positive series separates three costs: analytic truncation, polynomial hull overestimation, and conversion from nats to bits. For rational data the first two are rational bounds, not floating optimization diagnostics. High degree and many small cells can still be expensive, and no measured speedup is claimed. Better parameter-adaptive tails and a certified derivative enclosure could reduce work near an interior maximizer. Sharp scalar curvature adapted to γ, ν could improve the present uniform $4(1-\gamma)^2$ modulus. A complete equality classification for the zero-block endpoints is a separate research problem. Neither of these directions removes the need for analytic entropy and coding inputs.

7 Verification scope and conclusion

All ten inventoried public text files were read in fresh bounded literal displays, including both Hessian routes and all endpoint arguments. The two original private chapter bodies were also read freshly in full; this is not a whole-corpus reading. Pre-display SHA-256 captures and exact line intervals are preserved in the owned audit records, followed by post-read hashes and explicit unread complements. No private content was uploaded or used in public searches. Bibliography inclusion alone is not counted as reading an external proof. Source byte identity against the captured release trees is provenance, not proof validity or live status.

The independent owned executable checks target the endpoint coefficient identities, channel substitutions, branch linear-entropy coefficient, physical-coordinate transport, rational series and Bernstein enclosures. Numerical entropy diagnostics are separately identified; no finite grid is a theorem over all ranks, channels, or code lengths. The owned run passed 335 exact rational cases with 2,848 assertions: 159 thermal coefficient cases, 125 channel/linear-entropy cases, nine noncommuting block cases, 36 polynomial-hull/finite-tail cases, one worked certificate, and five physical-coordinate cases. Separately, nine float64 entropy diagnostics passed nine assertions at tolerance $2 \cdot 10^{-12}$; these are not interval proofs. The exact script used SymPy 1.11.1 and the diagnostics NumPy 1.23.5. Parent checks and prior reports are not included.

The demonstrated local increment is the explicit retained thermal slack and its conditional ensemble consequences, together with a source-specific positive-series certificate for the scalar problem. Classical strong concavity, physical metric transport, and Bernstein hulls are credited as tools. These results justify a notable local C3 contribution, not a claim of a new global capacity theorem or independent certification of the source's universal partner assertion.

References

[1] OpenAI, *Classical capacity and entropy inequalities for generalized amplitude damping*, September 24, 2026. Pinned public manuscript and source. Full assigned text read; the dependency boundary is stated in Section 1.

[2] S. Khatri, K. Sharma, M. M. Wilde, *Information-theoretic aspects of the generalized amplitude damping channel*, Phys. Rev. A 102, 012401 (2020), arXiv:1903.07747v2. Abstract/version metadata checked only.

[3] Z. Tang, C. Zhu, G. Bai, X. Wang, *Classical Capacity and Entanglement Cost of the Amplitude Damping Channel*, arXiv:2609.28592v1 (2026). Abstract/version metadata checked only.

[4] K. Fang, *The classical capacity of generalized amplitude-damping channels and its strong-converse exponent*, arXiv:2609.14621v2, October 6, 2026. Abstract/version metadata checked only; differs from the cached source's v1 citation.

[5] E. A. Carlen, *A remainder term for Hölder's inequality for matrices and quantum entropy inequalities*, arXiv:1605.04581v1 (2016). Abstract/version metadata checked; classical comparison, not an audited external proof.

[6] OpenAI Math repository, Lean scope declaration 276, public text read October 9, 2026. No Lean proof run or audit performed.

[7] M. Subbarao, original private RamCalBooks, volume 21, chapter body `ch01d-physical-metrics.tex`, lines 1–210. Freshly read physical-Gram, metric, and full-residual proofs; local confidential source.

[8] M. Subbarao, original private RamCalBooks, volume 16, chapter body `ch05d-clearance.tex`, lines 1–299. Freshly read Bernstein conversion, subdivision and remainder proofs; local confidential source, not an external quantum theorem.